OS TIERS

The OS Tiers architecture of the Monkey Head Project organizes Huey's operating system into three distinct but interconnected layers: MacroOS, MicroOS, and NanoOS. This layered structure optimizes strategic planning, operational execution, and real-time control.

MacroOS:
- Strategic layer that defines long-term objectives and allocates system-wide resources.
- Interfaces directly with governance structures to ensure compliance with constitutional rules.

MicroOS:
- Operational layer that translates strategic goals into actionable plans.
- Coordinates and manages subsystems to ensure efficiency and effectiveness.

NanoOS:
- Real-time control layer that manages low-level, immediate operations.
- Responds to environmental inputs, sensor data, and urgent system events.

Advantages:
- Scalability – Each layer can evolve independently as system demands grow.
- Resilience – Layered redundancy reduces the risk of catastrophic failures.
- Flexibility – Capable of adapting to new mission profiles and environmental conditions.

Governance Integration:
- Each layer logs actions and decisions in the Unified Audit Log for transparency.
- Governance approval is required for major changes to operational protocols at any layer.

Testing Objectives:
- Evaluate communication speed and reliability between OS tiers under various conditions.
- Measure performance gains from specialized task delegation.
- Assess recovery speed after simulated faults in each OS layer.